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Cumulative Reflection: A Reflective Journey

Through my education at Iowa State University, I have found that challenging problems cannot always be solved effectively from a single perspective or through our own experiences. Because of this, I have learned that the best way to solve problems is by considering multiple perspectives. These perspectives were built through the technical, personal, collaborative, and ethical experiences I have had throughout my engineering education.

The first perspective I have developed through my education is a strong theoretical foundation in engineering. This includes mathematics, physics, programming, computer systems, algorithms, and more. These foundations have prepared me to approach problems logically, break complex problems into smaller parts, and use technical knowledge to develop solutions.

Another perspective I added to my engineering toolbox is the pursuit of lifelong learning. Engineers must continually learn, even beyond the classroom, to prepare themselves to solve current and future problems. I know that I can solve problems that may be new to me. My experience at Iowa State has taught me how to learn new subjects and apply that knowledge to unfamiliar situations. I have learned that graduating does not mean I am finished with learning. Instead, graduating marks the beginning of a career where learning will continue to be an important part of being an engineer.

The third perspective I have learned during my time at Iowa State University is the importance of asking for help. Engineers can sometimes believe that their ideas are the best or

that they need to figure everything out themselves. However, the best engineers recognize the value of sharing knowledge with others and identifying when they do not know something. Throughout my education, I once held the belief that I should be able to face challenges on my own. Now that I understand the power of asking for help, I will use this skill in the future to help solve both engineering and non-engineering problems throughout my career.

Another experience that has shaped my engineering education is collaboration. Throughout my engineering courses, I have had many opportunities to work with partners and groups. Through these experiences, I have learned a lot; some have been very positive, while others have been more difficult. From my positive experiences, I have learned what effective teamwork looks like and strategies for fostering that type of environment. More difficult experiences have taught me how to work through difficult deadlines, disagreements, and different working styles. Over the course of my Iowa State journey, I have had the opportunity to work with many different people, which has brought new learning experiences and challenges. I have learned when to take a leadership role and when to step back and allow others to lead. I have also been challenged to work with many different types of personalities and find ways to succeed as a team. My dad once told me when I was young that working in groups is one of the most important learning opportunities in life. After my collaborative experiences at Iowa State, I would agree with that statement. These experiences have taught me how to navigate challenges within a team, and I am now confident in my ability to contribute effectively and work toward a successful outcome.

The final perspective I have gained while at Iowa State is the ethical perspective. When taking my engineering ethics course, my eyes were opened to the ethical challenges engineers can encounter. This was new to me because ethical dilemmas do not often occur within typical

coursework. In this course, I learned the importance of ethical decision-making and different strategies for handling ethical dilemmas that engineers may encounter throughout their careers. I learned that these situations may not always have a clear right answer and can involve competing responsibilities and consequences. Following this course, I feel better prepared to approach ethical dilemmas thoughtfully and consider my responsibilities as an engineer when making difficult decisions. More importantly, I understand that recognizing an ethical responsibility is only the first step. I must also be willing to act upon it and make decisions that uphold my responsibilities as an engineer.

My Iowa State engineering experience has prepared me to solve difficult challenges by taking advantage of the new perspectives I have gained throughout my education. These perspectives include engineering foundations, lifelong learning, the importance of asking for help, successful collaboration, and ethics. I am confident that by using these perspectives collectively, I can approach challenging problems from multiple angles and successfully develop solutions.

While at Iowa State, I expanded my knowledge horizons most by observing how those I look up to approach challenges. This may have meant going over difficult math problems with a respected professor. Rather than simply focusing on the answer, I wanted to understand how they approached the problem and what made them successful in solving it. Often, this meant learning to relax, trust that I knew what to do, and understand that if something does not work out, I should take a different approach.

I also talked to other classmates who were successful. I wanted to understand how they were able to digest the material, perform well on the exams, and complete the work. Through these interactions, I learned that hard work goes a long way. These students were not cheating the

system or relying on some secret strategy. They simply put a lot of time and effort into understanding the concepts so they could succeed in the class.

These experiences of observing successful individuals taught me that there is no secret to success and nobody is perfect. Rather, hard work, effort, and caring about what you do can help you learn difficult concepts and become more proficient in them.

I have also learned the importance of lifelong learning. At career fairs and through other interactions with industry professionals, I often ask the question, “What is the most important value successful people have in your industry?” The most common answer I received was the ability to learn. Some of the professionals even went on to say that very little of the actual content learned in college is directly applicable to your career. Instead, they highlighted that an engineering degree is more about learning how to learn and solve problems. This sentiment was further strengthened at my internship last summer. Experienced engineers within the organization shared their career journeys and explained how lifelong learning has given them the ability to continue developing and expand their career.

While spending time within a research group made up of undergraduate and graduate students led by an experienced professor in my department, the most important thing I learned was how to dive into new topics and spaces you are unfamiliar with and learn about them. An important part of this was learning to ask questions and acknowledge when I do not know something. I was surprised to see the professor leading the group ask so many questions and admit when he did not understand something. This experience taught me that experts in certain fields became experts through constant learning. Even experts are still learning and do not know everything.

These experiences have better equipped me to handle unforeseen challenges in my engineering career. I know that I will face problems that I have never seen before, but I am confident in my ability to learn and solve them. Lifelong learning has taught me that I do not need to know everything to be a successful engineer. A successful engineer is willing to learn, ask questions, and adapt to new challenges.

My college experience has illuminated the path for continuous skill enhancement in the face of ever-evolving problems through the foundational skills I have gained and my understanding of the importance of continuous learning. I know that whatever problem I face, the combination of the skills I already have and my ability to learn and tackle new problems will give me the ability to solve them.

I have harnessed these skills in a group project during my junior year. We were building an application using programs we were unfamiliar with. At times, I did not really know the path to our goal, but I knew I had to start somewhere. Using my prior programming skills and ability to learn, we achieved our goal.

This project also challenged my leadership skills. While I am comfortable using my engineering skills to tackle technical problems, leadership and collaboration proved to be more difficult. I still used my ability to learn to improve my leadership skills, but it also took collaboration and learning from my own mistakes before I became comfortable in my leadership role. This experience showed me that adapting to new challenges is not limited to technical skills. It also requires me to continuously develop how I work with and lead others.

Looking back on my journey, I would have taken more advantage of the resources at Iowa State University. The people and resources across all colleges and departments are endless. I wish I had used more of these resources to help me grow academically, professionally, and

personally. I have only taken advantage of some of these resources in the later part of my education and have found them very useful.

Overall, I am very pleased with my educational journey and would not change much. I believe that the mistakes and challenges I faced have helped me grow into the person I am today. Because of that, if I made my journey easier, I probably would not have become as developed. The challenges I faced were an important part of my growth, and I would not want to take those experiences away from my journey.

A recent engineering topic I have embraced was during an internship experience. At this internship, I had to learn embedded machinery programming in MATLAB Simulink within the company's structure. The process of grasping this concept began with starting somewhere and getting my feet wet. I learn best when I run into problems and find a way to solve them. This may be on my own or by asking others who are more experienced than me. Once I felt comfortable with one area within this concept, I would strive to find another area that I am still unfamiliar with or one that built on top of the previously learned topic.

This approach worked well for me. I progressed quickly and walked away much more experienced and knowledgeable in the engineering concept. A challenge I faced was making assumptions about a process because I had worked with something similar in the past. When stepping into unknown concepts, I learned you cannot assume they will work exactly like a concept you have already mastered. For example, while programming a certain feature, I thought I was going above and beyond by solving what I perceived as problems outside the scope of my assignment. What I did not understand was that although they appeared to be problems, the structure of the program meant that it was working correctly, and my additional work was unnecessary.

In my past experiences with similar concepts, this behavior needed to be fixed. However, in this new environment, it was different. I learned that I should not approach a new topic with assumptions based solely on my prior knowledge. Instead, I need to understand how the new system works before deciding what is or is not a problem. This experience helped me become more comfortable learning unfamiliar concepts and adapting my existing knowledge to new situations.

A moment where I have, and continue, to translate theoretical knowledge into practical innovation is through my senior design project. Our client is a business student with a motivation to decrease food waste. To do so, our goal is to create a device that can inventory food within a refrigerator and track the freshness of each item so consumers can use food before it goes bad, reducing wasted food, along with many other features. The challenge the client has faced is that they do not have the technical knowledge to translate this goal into a physical product. This is why a small group of engineers and I were approached with this idea.

Using our theoretical knowledge, we can take this innovative idea and transform it into a real, working device. Since this is an ongoing project, we do not have a final outcome yet, but our progress has been good. This experience has given me a greater sense that my foundational engineering skills give me the ability to solve almost any problem I encounter. I have learned that there is no reason to settle for poor-quality work because I have the knowledge to either fix a problem or learn how to fix it. This mindset has helped me approach challenges with more confidence and creativity.

While at Iowa State, a strategy that I have learned to use when conquering complex engineering concepts is persistence and believing in myself. This strategy has been shaped through hours of debugging in programming courses. While trying to fix errors in programming

courses, it can be very frustrating and almost hopeless at times. Despite this, I always figured the problem out and got past the hardship. Unconsciously, I think this made it harder for me to get frustrated when I did not understand a concept or made mistakes. This is because I have proved to myself that I can work through difficult situations and find success.

Another strategy I have learned is the importance of taking a step back and slowing down. At times when I am struggling to understand a concept, I have found that getting my mind off the frustration point helps me regain my focus. Sometimes this may mean taking a short walk or getting something to eat. When I return to the work that was challenging me, I make sure to slow down and be gentle with myself because I am learning.

The biggest factors driving these adaptations have been my academic need to understand concepts and complete projects for my grades, as well as my own mental well-being. I learned to become more mentally resilient when facing difficult challenges. I knew it was unhealthy to become frustrated when faced with challenges. Now that I use these strategies, I no longer become as frustrated when things do not work, and I know how to regain my focus. This has made it easier for me to learn new concepts and work through difficult engineering problems.

An area where I could grow further is leadership. Although I have gained leadership skills at Iowa State University, I have not experienced managing multiple individuals who directly report to me. I believe that when I get that opportunity, I will reflect on the foundational skills and strategies I have learned throughout my educational journey to guide me in the right direction as a manager.

I think of my learned knowledge as a large toolbox. As I learn new things, I can add them to my toolbox, giving me more tools to use when solving problems. To stay motivated to learn, I will challenge myself to complete tasks that may be unfamiliar to me. Through my experiences

as an engineer, I know that the toughest challenges are excellent learning opportunities. In the future, I hope to be in a position where I am motivated and rewarded for continuous learning and professional growth while continuing to use the skills and perspectives I have developed throughout my time at Iowa State University.